

Bacteria and Plant Nutrition

A variety of soil bacteria play roles in plant nutrition. Some engage in mutually beneficial chemical exchanges with plant roots. Others enhance the decomposition of organic materials and increase nutrient availability.

Rhizobacteria

Rhizobacteria are bacteria that live either in close association with plant roots or in the **rhizosphere**, the soil closely surrounding plant roots. Many rhizobacteria form mutually beneficial associations with plant roots. Rhizobacteria depend on nutrients such as sugars, amino acids, and organic acids that are secreted by plant cells. Up to 20% of a plant's photosynthetic production may be used to fuel these complex bacterial communities. In return, plants reap many benefits from these mutualistic associations. Some rhizobacteria

produce antibiotics that protect roots from disease. Others absorb toxic metals or make nutrients more available to roots. Still others convert gaseous nitrogen into forms usable by the plant or produce chemicals that stimulate plant growth. Inoculation of seeds with plant-growth-promoting rhizobacteria can increase crop yield and reduce the need for fertilizers and pesticides.

Some rhizobacteria are free-living in the rhizosphere, whereas others are **endophytes** that live between cells within the plant. Both the intercellular spaces occupied by endophytic bacteria and the rhizosphere associated with each plant root system contain a unique and complex cocktail of root secretions and microbial products that differ from those of the surrounding soil. A recent study revealed that the compositions of bacterial communities living endophytically and in the rhizosphere are not identical (**Figure 37.10**). A better

▼ Figure 37.10 Inquiry

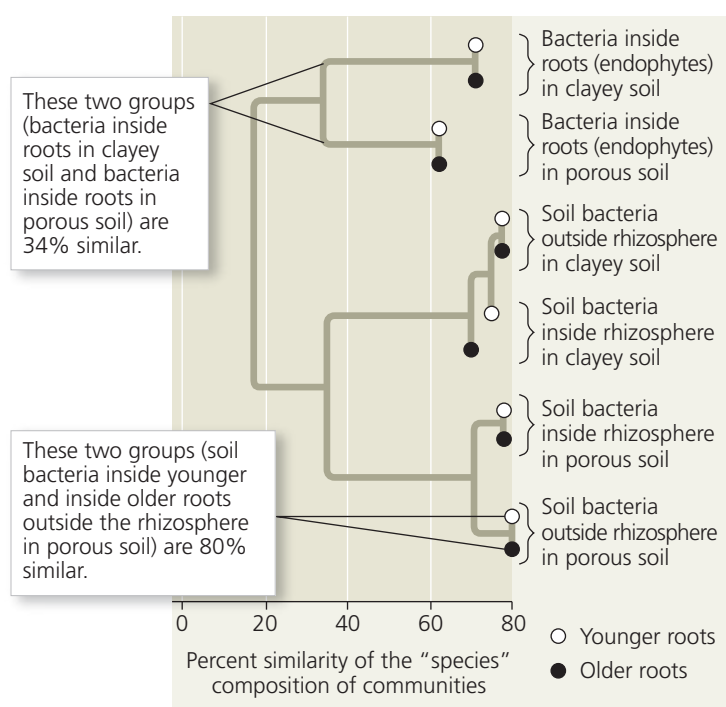
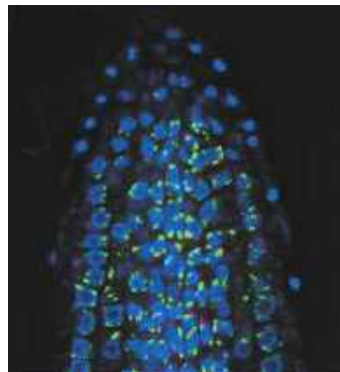
How variable are the compositions of bacterial communities inside and outside of roots?

Experiment The bacterial communities found within and immediately outside of root systems are known to improve plant growth. In order to devise agricultural strategies to increase the benefits of these bacterial communities, it is necessary to determine how complex they are and what factors affect their composition. A problem with studying these bacterial communities is that a handful of soil contains as many as 10,000 types of bacteria, more than all the bacterial species that have been described. Researchers cannot simply culture each species and use a taxonomic key to identify them; a molecular approach is needed.

Jeffery Dangel and his colleagues estimated the number of bacterial "species" in samples using a technique called *metagenomics* (see Concept 21.1). The bacterial community samples differed in location (endophytic, inside the rhizosphere, or outside the rhizosphere), soil type (having large amounts of clay or porous), and the developmental stage of the root system with which they were associated (old or young). The DNA from each sample was purified, and the polymerase chain reaction (PCR) was used to amplify the DNA that codes for the 16S ribosomal RNA subunits. Many thousands of DNA sequence variations were found in each sample. The researchers lumped the sequences that were more than 97% identical into "taxonomic units" or "species." (The word *species* is in quotation marks because "two organisms having a single gene that is more than 97% identical" is not explicit in any definition of species.) Having established the types of "species" in each community, the researchers made a tree diagram showing the percent of bacterial "species" found in common in each community.

Results This tree diagram breaks down the relatedness of bacterial communities into finer and finer levels of detail.

▼ Bacteria (green) on surface of root (fluorescent LM)



Data from D. S. Lundberg et al., Defining the core *Arabidopsis thaliana* root microbiome, *Nature* 488:86–94 (2012).

The two explanatory labels give examples of how to interpret the diagram.

Conclusion The "species" composition of the bacterial communities varied markedly according to the location inside the root versus outside the root and according to soil type.

INTERPRET THE DATA (a) Which of the three community locations was least like the other two? (b) Rank the three variables (community location, developmental stage of roots, and soil type) in terms of how strongly they affect the "species" composition of the bacterial communities.